

Exercise Sheet 2 System Safety

Exercise 2.1: Safety Vocabulary

optional

Define the following terms in your own words and give a concrete example from an autonomous driving context for each:

1. **Loss**
2. **Hazard**
3. **Risk**

1. **Loss:**

A loss is an adverse outcome involving harm to people, property, or the environment that must be prevented.

Example: Vehicle hits a pedestrian → injury or death.

2. **Hazard:**

A hazard is a system state or condition that, under certain circumstances, can lead to a loss.

Simple understanding: Dangerous situation that can cause a loss.

Example: Vehicle is moving while a pedestrian is in its path. (Important: accident has not happened yet, but it can happen.)

3. **Risk:**

Risk is the combination of the likelihood of a hazard occurring and the severity of the resulting loss.

Simple understanding: How likely something bad is + how bad it would be.

Example: Missing pedestrian detection

Likelihood: medium

Severity: very high

→ high risk

Loss → outcome (damage)

Hazard → situation (danger)

Risk → evaluation (likelihood + severity)

STPA: System-Theoretic Process Analysis

STPA Step 1: Define Purpose and Identify Losses, Hazards, and ODD

Exercise 2.2: ODD Specification

Based *only* on the system description in the course Roadmap (you have not yet seen the training data), define the **Operational Design Domain** (ODD) for the CARLA system. Cover at least the following dimensions: weather, lighting, camera condition, scene type, and vehicle speed.

For each dimension, state:

1. The **operating conditions** (where the system is considered valid).
2. The **non-operating conditions** (where it is not).
3. How an ODD violation could be **detected at runtime**.

Hint: You are defining the ODD *before* looking at the data. In Exercise Sheet 3, you will compare this ODD to the actual training data and (possibly) discover gaps.

1. ODD:

The Operational Design Domain (ODD) is the **set of conditions under which the system is intended to operate safely**. Outside this domain, no safety guarantees apply. [simple exp: Where the system is allowed to work.]

2. **CARLA** = a simulated world to train and test self-driving systems.

3. Dimensions:

1. Weather

- **Operating:** clear, sunny, light clouds
- **Non-operating:** heavy rain, fog, snow
- **Detection of violation:**
use camera visibility checks (blur, low contrast) or weather sensors

2. Lighting

- **Operating:** daytime
- **Non-operating:** night, low-light condition.
- **Detection:**
measure brightness of image (pixel intensity)

3. Camera condition

- **Operating:** clean, fixed, forward-facing
- **Non-operating:** dirty, occluded, misaligned

- **Detection:** detect blur, obstruction, or abnormal image patterns

4. Scene type

- **Operating:** urban environment (roads, intersections)
- **Non-operating:** highways, rural roads, unusual environments
- **Detection:**
scene classification model or map-based verification

5. Vehicle speed

- **Operating:** low speed (e.g., city/intersection driving)
- **Non-operating:** high-speed driving
- **Detection:**
check speed sensor values

Main idea:

- Inside ODD → system is expected to work
- Outside ODD → behavior is unreliable
- **System must detect ODD violations**

Exercise 2.3: Losses

STPA begins by identifying the **losses** that must be prevented - unacceptable outcomes involving harm to people, property, or the environment.

For each loss, identify its ID (L-1, L-2, ...), briefly describe it, and explain why it is unacceptable:

ID	Loss Description	Why Unacceptable
L-1	Loss of vehicle control leading to unsafe behavior	Can lead to accidents involving multiple participants.
L-2	Collision causing damage to vehicles or infrastructure	Financial damage and possible secondary risks
L-3	Injury or death of a pedestrian	Direct harm to human life
L-4	Violation of traffic laws	Legal consequences and unsafe system behavior

L-5	Injury to passengers inside the vehicle	The system must also ensure safety of its own occupants
-----	---	---

Exercise 9.8: Extending the Safety Analysis for OOD

Exercise 2.4: Hazards

A **hazard** is a system state that, combined with worst-case environmental conditions, can lead to a loss. Hazards are properties of the *system*, not of individual components.

1. Identify at least four system-level hazards (H-1, H-2, ...) for the CARLA system.
2. For each hazard, state which loss(es) from Exercise 2.3 it can lead to.
3. Rate each hazard's likelihood (low / medium / high) and its severity (low / medium / high).

Fill in the following table:

ID	Hazard	Loss(es)	Likelihood	Severity
H-1 [ref in exc 9]	Vehicle is moving while a pedestrian is in its path. i.e., Vehicle fails to brake for a pedestrian	L-3	Medium	High
H-2	Vehicle fails to detect another vehicle or obstacle in front	L-2, L-1	Medium	High
H-3	Vehicle continues driving despite unsafe or unknown environment (outside ODD)	L-1, L-2, L-3	High	High
H-3 revised	Vehicle continues driving despite operating on OOD inputs [fog/night images] that are not detected, leading to unreliable perception outputs.	L-1, L-2, L-3	High	High
H-4	Planner issues incorrect control action (eg: accelerates instead of braking)	L-1, L-2, L-5	Medium	High

H-5	Traffic light is detected as present, but its state (red/green) is unknown, leading to unsafe decisions at intersections	L-2, L-3, L-4	Medium	High
H-6 [exc 8]	The vehicle continues driving despite adversarially perturbed inputs that cause incorrect perception outputs (eg: missed pedestrian, vehicle, or traffic light).	L-1, L-2, L-3, L-4	Medium	High
	Example: A pedestrian is present, but the pedestrian classifier is fooled by an FGSM attack and outputs a false negative. The planner continues driving instead of braking.			

STPA Step 2: Model the Control Structure

Exercise 2.5: Control Structure Diagram

Draw the control structure for the CARLA system. Your diagram must include:

- All components listed in the system description (camera, perception models, distance oracle, planning module, actuators, human operator).
- **Control actions** (solid arrows): commands flowing from controllers to controlled processes (e.g., “brake command”).
- **Feedback** (dashed arrows): information flowing back (e.g., “perception labels”, “distance estimates”, “vehicle speed”).

Hint: There are at least two controllers in this system: the planning module and the human operator. The perception models and distance oracle should be considered *sensor/feedback*, not controllers themselves - they provide information to the controllers.

Hint: The traffic-light model only detects presence, not traffic light state. Consider what decisions the planner can (or cannot) safely make with this information.

1. Control Structure:

A control structure represents the system as **controllers, controlled**

processes, control actions, and feedback loops.

It shows:

- who makes decisions
- what actions are taken
- what information is used

2. Diagram:

STPA Step 3: Identify Unsafe Control Actions and Derive Safety Constraints

Exercise 2.6: Unsafe Control Actions (UCAs)

Using your control structure, identify **unsafe control actions** for at least **two different controllers**. For each UCA, also indicate which hazard(s) from Exercise 2.4 it could cause or lead to (reference them by ID, e.g., “H-1”, “H-2”).

For each controller, consider all four UCA types:

1. **Not provided** - a required action is missing.
2. **Provided unsafely** - the action is issued when it should not be.
3. **Wrong timing / order** - too early, too late, or out of sequence.
4. **Wrong duration** - applied too long or stopped too early.

Fill in the table below (add rows as needed):

ID	Controller	Control action	UCA type	Hazard(s)	Unsafe scenario
UCA-1 [ref in exc 9]	Planning module	Brake command	-----	H1, H2	Planner does not issue braking when a pedestrian or obstacle is in the path.
UCA-2	Planning module	Continue command	Provided unsafely	H-1, H-2, H-3	Planner continues driving despite unsafe environment or obstacle

UCA-3	Planning module	Brake command	Wrong timing	H1, H2	Brake is applied too late to avoid collision
UCA-4	Planning module	Brake command	Wrong duration	H1, H2	Brake is released too early before stopping
UCA-5	Human operator	Override command	----	H-3, H-4	Human does not intervene when system behaves unsafely
UCA-6	Human operator	Override command	Provided unsafely	H-2, H-4	Human overrides correct system decision and causes unsafe action
UCA-7	Human operator	Override command	Wrong timing	H-1, H-2	Human intervenes too late to prevent accident
UCA-8	Human operator	Override command	Wrong duration	H-4	Human releases or applies control for incorrect duration
UCA-9	Planning module	Continue command	Provided unsafely	H-1, H-2, H-3	Planner continues driving while perception inputs are OOD and the resulting perception outputs are unreliable because OOD was not detected.
					The planner keeps driving even though the camera sees

					fog/night images that make the perception model unreliable.
UCA-10 [exc 8]	Planning module	Continue command	Provided unsafely	H-1, H-2, H-6	Planner continues driving while perception outputs are incorrect due to an adversarially perturbed input that was not detected.

Exercise 2.7: Safety Constraints

For each UCA from Exercise 2.6, derive at least one **safety constraint** - a requirement that, if satisfied, prevents the UCA from occurring or limits its consequences.

Classify each constraint as either:

- **Model-level** - can be verified by testing or evaluating the corresponding perception model (this is what you will do in Sheets 4–9).
- **System-level** - requires a design measure beyond the model (e.g., redundancy, operator protocol, fallback behavior).

UCA	Safety constraint	Level	Verification
UCA-1	The planner must issue a braking command when a pedestrian or obstacle is within stopping distance	System-level	Scenario-based testing / simulation
UCA-2	The planner must not continue	System-level	ODD violation tests /

	driving when the environment is unsafe or uncertain		safety logic checks
UCA-3	The system must ensure braking is initiated within a safe time threshold	System-level	Latency measurement / timing tests
UCA-4	The braking action must be maintained until the vehicle reaches a safe stop	System-level	Simulation tests with stopping distance
UCA-5	The system must alert the human operator when unsafe conditions occur	System-level	Human-in-the-loop testing
UCA-6	The system must prevent unsafe overrides or provide warnings to the operator	System-level	Override safety checks / interface testing
UCA-7	When an OOD condition is detected, the vehicle shall transition to a minimal-risk state (e.g. slow down, stop, or request human intervention).	System-level	Simulation-based OOD response testing
UCA-7.1	The OOD monitor shall detect and flag OOD inputs with sufficiently high detection performance before perception outputs are used for planning.	Model-level	OOD evaluation (e.g, AUROC using fog/night data)
UCA-8 [exc 8]	The perception models shall maintain acceptable recall under adversarial perturbations up to a specified ϵ budget. Recall drop shall remain below a predefined threshold (eg: <20% for $\epsilon \leq 0.05$).	Model-level	FGSM robustness evaluation
UCA-8.1	When a prediction has low	System-level	Simulation-based

[exc 8]	confidence or appears anomalous, the planner shall transition to a minimal-risk state (slow down, stop, or request human intervention).		safety testing
UCA-9 [exc-7.7 (q2)]	The pedestrian classifier shall maintain an ECE below 0.25 on the validation set. [my results]: The calibrated pedestrian model achieved an ECE of 0.427 [after temp scaling] and therefore does not satisfy the constraint.	Model-Level	ECE evaluation before and after temperature scaling.
UCA-9.1	The planner shall use the cost-optimal threshold $\tau = 0.0099^*$ for pedestrian detection and transition to a minimal-risk state when prediction confidence is low.	System-Level	Simulation-based safety testing and threshold evaluation. [so we check in simulation whether the planner behaves safely when using the new threshold.] [we just compare safety performance (eg: missed pedestrians, collisions, total cost) using $\tau = 0.5$ vs $\tau^* = 0.0099$.]

STPA Step 4: Identify Causal Loss Scenarios

Exercise 2.8: Causal Loss Scenarios

For each UCA from Exercise 2.6, identify **at least one causal scenario** - a specific mechanism or sequence of events that could cause the UCA to occur in practice.

For each scenario, describe:

1. **The causal factor:** What component failure, misunderstanding, or adverse condition triggers the UCA?
2. **Why it occurs:** What is the root cause or underlying reason?
3. **Connection to constraint:** Which safety constraint from Exercise 2.7 is designed to prevent or mitigate this scenario?

Hint: Consider the following causal mechanisms (from the lecture):

- **Incorrect feedback:** A sensor or perception model provides wrong information to the controller.
- **Flawed internal model:** The controller misinterprets correct information (e.g., low confidence treated as “no detection”).
- **Timing delay:** Information arrives too late for the controller to respond safely.
- **Human factor:** The human operator fails to intervene due to attention lapse, delayed reaction, or mode confusion.

UCA	Causal scenario	Root cause	Related constraint
UCA-1	Pedestrian is present but not detected and the planner does not issue brake	Perception model misses pedestrian	Planner must issue braking command when obstacle is within stopping distance
UCA-2	System is outside ODD (eg: night or low light conditions), but planner continues driving	No ODD violation detection	Planner must not continue driving in unsafe or unknown conditions
UCA-3	Detection arrives late and the braking command issued too late	High latency in perception or processing	System must ensure braking within safe time threshold
UCA-4	Vehicle starts braking but stops too early and	Incorrect distance estimation or	Braking must continue until safe

	hits obstacle	control logic	stop is reached
UCA-5	System behaves unsafely but human does not intervene	Human inattention or lack of alert	System must alert operator in unsafe situations
UCA-6	Human reacts too late to prevent collision	Delayed perception or slow reaction time	System must provide timely alerts for intervention
UCA-7 [exc9 7.7]	Pedestrian is present, but the classifier outputs a false negative with high confidence and the planner does not issue a brake command.	Overconfident misclassification due to poor model calibration	The perception model shall maintain acceptable calibration and the planner shall use confidence-aware fallback behavior. [slow down, stop, or request human intervention when prediction confidence is low or uncertain.]
UCA-8 [exc9]	Pedestrian is present, but the planner uses the default decision threshold ($\tau = 0.5$) and therefore classifies the pedestrian as absent, causing the vehicle to continue driving.	Inappropriate decision threshold that does not account for the high cost of missed pedestrians.	The planner shall use the cost-optimal threshold ($\tau^* = 0.0099$) for pedestrian detection instead of the default threshold ($\tau = 0.5$).